Mathematics for Computing

Assignment 1

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Contents

1 10 Sets in Mathematical Form

Below are 10 standard examples of sets represented in mathematical notation:

1. **Natural Numbers:** $\mathbb{N} = \{1, 2, 3, 4, \dots\}$
2. **Whole Numbers:** $\mathbb{W} = \{0, 1, 2, 3, \dots\}$
3. **Integers:** $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
4. **Rational Numbers:** $\mathbb{Q} = \left\{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\right\}$
5. **Real Numbers:** $\mathbb{R} = \{x \mid x \text{ is a rational or irrational number}\}$
6. **Set of Even Integers:** $E = \{2n \mid n \in \mathbb{Z}\}$
7. **Set of Odd Integers:** $O = \{2n + 1 \mid n \in \mathbb{Z}\}$
8. **Set of Prime Numbers:** $P = \{x \in \mathbb{N} \mid x > 1 \text{ and the only divisors of } x \text{ are 1 and } x\}$
9. **Boolean Set:** $B = \{0, 1\}$
10. **A Continuous Interval:** $I = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$

2 General Form of the Inclusion-Exclusion Principle

The Inclusion-Exclusion Principle is a counting technique used to calculate the number of elements in a union of multiple sets by systematically adding and subtracting the sizes of their intersections.

For a finite number of sets $A_1, A_2, \dots, A_n$, the general mathematical form is given by:

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1}^{n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| \\ - \cdots + (-1)^{n-1} |A_1 \cap A_2 \cap \cdots \cap A_n|$$

3 Types of Sets with AI and Data Science Examples

- **Finite Set:** A set containing a countable number of elements.
  Example (AI/DS): The set of predefined object classes detected by a YOLOv8 waste classification model deployed on an edge device: $W = \{\text{plastic}, \text{glass}, \text{metal}, \text{paper}\}$.

- **Infinite Set:** A set that contains an endless number of elements.
  Example (AI/DS): The set of all possible continuous weight and bias updates generated during the training loop of a Generative Adversarial Network (GAN).

- **Empty (Null) Set ($\emptyset$):** A set containing no elements.
  Example (AI/DS): The set of false positive bounding boxes generated by an object detection model that achieves perfect 100% precision on a given test batch.

- **Subset ($\subseteq$):** Set A is a subset of B if every element of A is also in B.
  Example (AI/DS): The set of images containing solely organic waste is a subset of the universal set of all waste images captured by an autonomous surface water cleaning vehicle.

- **Disjoint Sets:** Sets that have no elements in common (their intersection is the empty set).
  Example (AI/DS): In a binary classification dataset, the set of *Training Data* and the set of *Validation Data* must be perfectly disjoint to prevent data leakage.

4 Partially Ordered Sets (POSET) and Hasse Diagrams

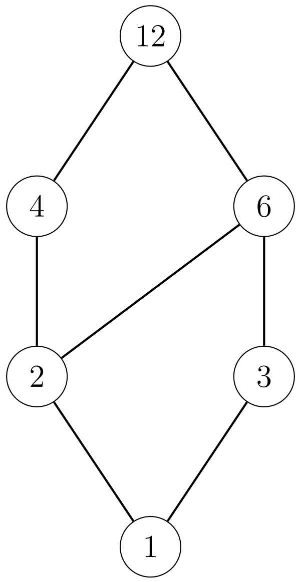
A Partially Ordered Set (POSET) is a set combined with a relation $\preceq$ that specifies a partial order. To be a partial order, the relation must be:

1. **Reflexive:** $a \preceq a$ for all a.
2. **Antisymmetric:** If $a \preceq b$ and $b \preceq a$, then $a = b$.
3. **Transitive:** If $a \preceq b$ and $b \preceq c$, then $a \preceq c$.

A Hasse diagram is a graphical rendering of a POSET where elements are nodes, and edges represent the relation, drawn upwards to imply direction (removing self-loops and transitive edges for clarity).

4.1 Example 1: The Divisibility Relation

Consider the set $A = \{1, 2, 3, 4, 6, 12\}$ under the relation "divides" $(a \mid b)$.



4.2 Example 2: The Subset Relation on a Power Set

Consider the set $S = \{a, b, c\}$. Let the POSET be the power set of S, $\mathcal{P}(S)$, ordered by the subset relation $\subseteq$.